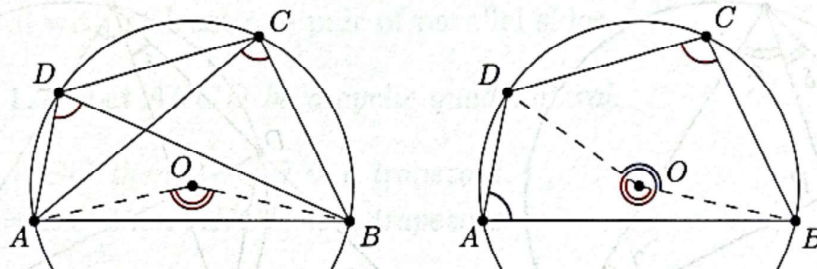


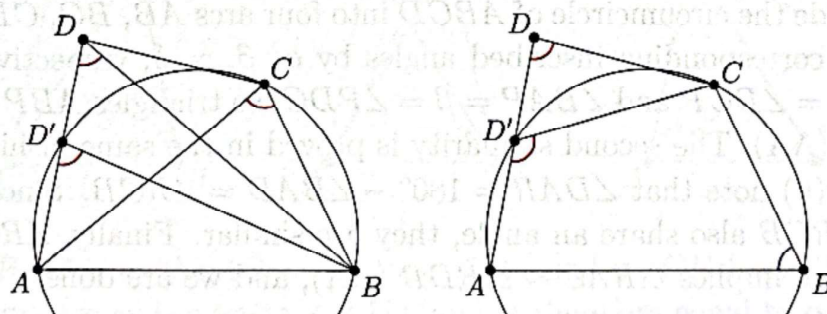
Proposition 1.29 (The key properties of cyclic quadrilaterals). *Let $ABCD$ be a convex quadrilateral. Then:*

- (a) *If $ABCD$ is cyclic then any of its sides is visible from the other two vertices under the same angle, and any of its diagonals is visible from the other two vertices under angles that sum up to 180° .*
- (b) *If there is a side of $ABCD$ that is visible from the other two vertices under the same angle, then $ABCD$ is cyclic.*
- (c) *If there is a diagonal of $ABCD$ that is visible from the other two vertices under the angles that sum up to 180° , then $ABCD$ is cyclic.*

Proof. For (a), denote by O the circumcenter of $ABCD$. By the Inscribed Angle Theorem $\angle ACB = \frac{1}{2}\angle AOB = \angle ADB$, and the other three equalities are proved similarly. Furthermore, as angles by O determined by segments OB and OD sum up to 360° , angles by A and C add up to $\frac{1}{2}360^\circ = 180^\circ$. Analogously, we obtain $\angle B + \angle D = 180^\circ$.



In (b), let us without loss of generality assume $\angle ACB = \angle ADB$, and denote by D' the second intersection of line AD and the circumcircle of triangle ABC . By (a), $\angle AD'B = \angle ACB = \angle ADB$ implying that lines DB and $D'B$ are parallel. Thus points D and D' coincide and quadrilateral $ABCD$ is cyclic.



In (c) we proceed similarly. Assume that $\angle ABC + \angle CDA = 180^\circ$ and denote by D' the second intersection of line AD and circumcircle of triangle ABC . Again by (a), $\angle AD'C = 180^\circ - \angle ABC = \angle ADC$, so $CD \parallel CD'$, $D = D'$, and $ABCD$ is cyclic. $\square$

As a direct consequence of the previous proposition we obtain that given a circle ω , its fixed chord AB , and a variable point X on ω , there are only two

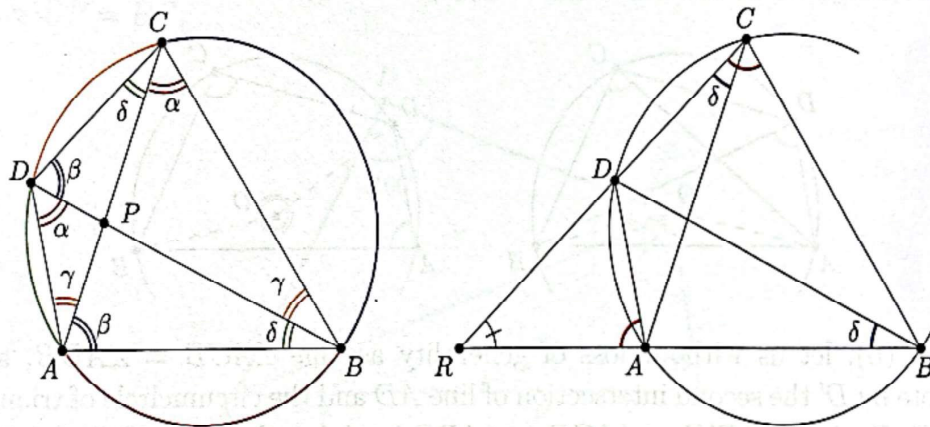
possible magnitudes of $\angle AXB$ depending on the relative position of line AB and point X . Moreover, these two magnitudes add up to 180° .

On the other hand, given a segment AB and angle φ , the locus of points X for which $\angle AXB = \varphi$ consists of two circular arcs symmetric with respect to AB .

Also in a configuration with four points on a circle one can find many similar triangles.

Corollary 1.30. *Let $ABCD$ be a cyclic quadrilateral, let P be the intersection of its diagonals and let R be the intersection of rays BA and CD . Then:*

- (a) $\triangle ABP \sim \triangle DCP$ and $\triangle BCP \sim \triangle ADP$,
- (b) $\triangle RAD \sim \triangle RCB$ and $\triangle RAC \sim \triangle RDB$.



Proof. Divide the circumcircle of $ABCD$ into four arcs AB , BC , CD , DA and denote the corresponding inscribed angles by α , β , γ , δ , respectively. Then $\angle PBA = \delta = \angle DCP$ and $\angle BAP = \beta = \angle PDC$, so triangles ABP and DCP are similar (AA). The second similarity is proved in the same fashion.

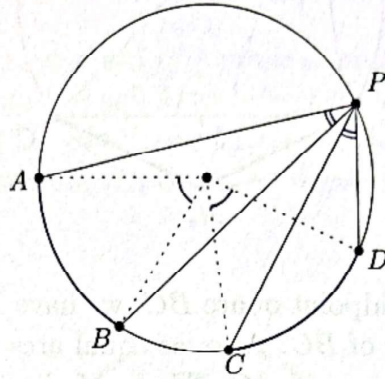
In part (b) note that $\angle DAR = 180^\circ - \angle BAD = \angle RCB$. Since triangles RAD and RCB also share an angle, they are similar. Finally, $\angle RCA = \delta = \angle DBR$ which implies $\triangle RAC \sim \triangle RDB$ (AA), and we are done. $\square$

Note that the triangles are similar indirectly. We will discuss this further in the section concerning antiparallelism.

Now, we establish an important corollary of the Inscribed Angle Theorem.

Corollary 1.31 (Correspondence between arcs and angles). *Let AB and CD be equal arcs of a circle ω . Then the inscribed angles corresponding to these arcs are equal.*

Proof. As the arcs are equal, they occupy the same portion of the perimeter of ω and the corresponding central angles are equal. Therefore the corresponding inscribed angles are also equal. $\square$



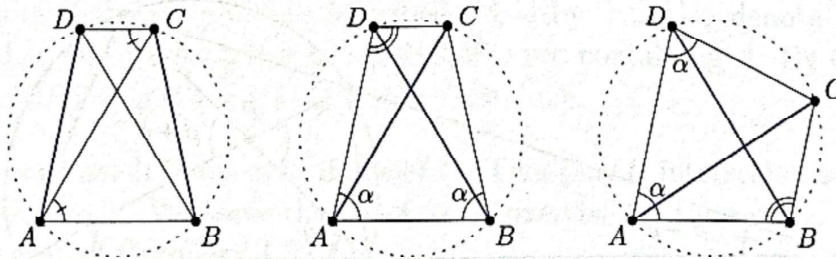
We illustrate this corollary by two examples. Recall that trapezoid is a quadrilateral with at least one pair of parallel sides.

Example 1.7. Let $ABCD$ be a cyclic quadrilateral.

(a) If $AD = BC$ then $ABCD$ is a trapezoid.

(b) If $AC = BD$ then $ABCD$ is a trapezoid.

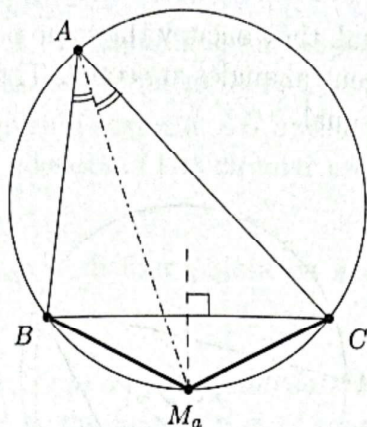
Proof. (a) From $AD = BC$ we infer that the minor arcs AD and BC are equal implying that $\angle DCA$ and $\angle BAC$ are equal. Hence $AB \parallel CD$ and $ABCD$ is a trapezoid.



(b) Let $\angle BAD = \alpha$. As $AC = BD$, one of the angles $\angle CBA$, $\angle ADC$ subtends the same arc as the angle $\angle BAD$, and is therefore equal to α too.

If $\angle CBA = \angle BAD = \alpha$ then $\angle ADC = 180^\circ - \angle CBA = 180^\circ - \alpha$ and the lines AB and CD are parallel. Similarly, if $\angle ADC = \angle BAD = \alpha$ then $\angle CBA = 180^\circ - \alpha$ and AD and BC are parallel. $\square$

Example 1.8. Let ABC be a triangle inscribed in a circle ω and let M_a be the midpoint of arc BC of ω that does not contain point A . Then the internal angle bisector of $\angle A$ and the perpendicular bisector of BC pass through M_a .



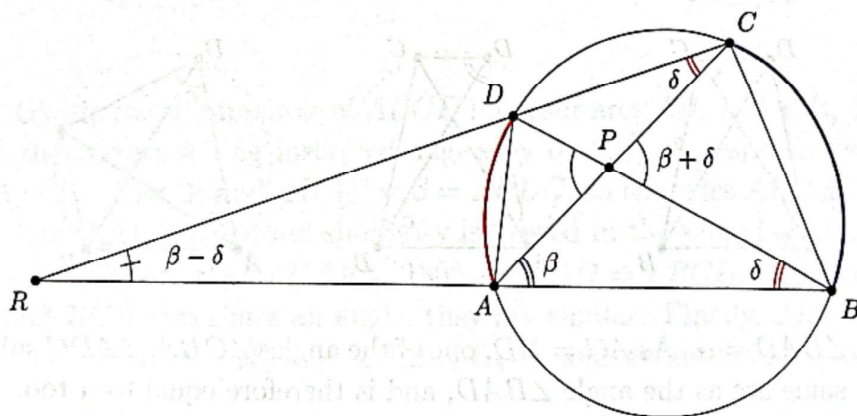
Proof. Since M_a is the midpoint of arc BC , we have $BM_a = CM_a$, so M_a lies on perpendicular bisector of BC . Also, as equal arcs are intercepted by equal angles, we have $\angle BAM_a = \angle M_aAC$. Thus, M_a lies on the angle bisector of $\angle A$. $\square$

The following corollary makes angle-chasing in cyclic quadrilateral easier.

Corollary 1.32. Let $ABCD$ be a quadrilateral inscribed in a circle ω and denote by P the intersection of its diagonals. Suppose that rays BA and CD intersect at R . Finally, denote the inscribed angles corresponding to arcs BC , DA (not containing A, B) by β, δ . Then

(a) $\angle BPC = \beta + \delta$,

(b) $\angle BRC = \beta - \delta$.



Proof. For part (a) using the property of an exterior angle in triangle ABP we get

$$\angle BPC = \angle BAP + \angle PBA = \beta + \delta.$$

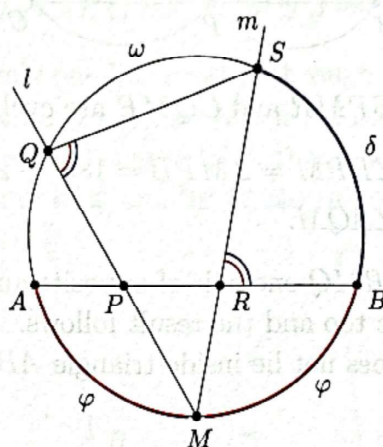
Similarly in part (b), looking at triangle ACR we obtain

$$\angle BRC = \angle BAC - \angle RCA = \beta - \delta.$$

Thus, the corollary is proven. $\square$

This corollary in fact asserts that we can express angle between two chords of a circle in terms of inscribed angles corresponding to some arcs of that circle. With this result we could serve the following proof without words!

Example 1.9. Let AB be a chord of a circle ω and let M be the midpoint of arc AB . Let line l passing through M intersect the chord AB at P and circle ω for the second time at Q . Similarly, let line m ($m \neq l$) passing through M intersect the chord AB and circle ω at R, S , respectively. Show that points P, Q, R, S lie on a circle.



Proof. It is enough to prove that $\angle PQS + \angle SRP = 180^\circ$ or equivalently $\angle MQS = \angle BRS$. Observe that shorter arcs MA and MB of ω are equal, and denote the corresponding inscribed angle by φ . Also denote by δ the inscribed angle corresponding to arc BS of ω not containing A . By Corollary 1.32(a) $\angle BRS = \varphi + \delta = \angle MQS$, so we are done. $\square$

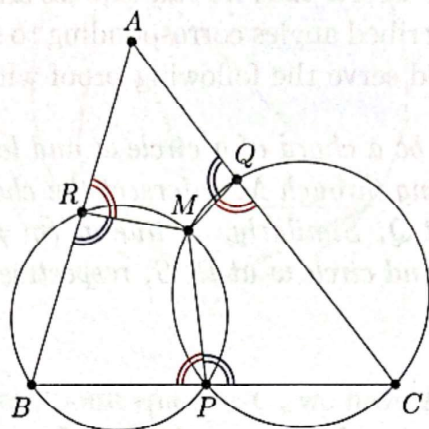
The same result holds even if one of the lines (say l) intersects the line AB outside the circle. We leave the proof as an exercise for the reader, this time with the help of Corollary 1.32(b).

If one wants to prove that three curves pass through a common point, it is often convenient to define the intersection of two of these curves and show that it lies on the third one (a method we have already seen in use in the proofs of the Propositions 1.6 and 1.7).

Theorem 1.33 (Miquel's⁶ pivot theorem). Let P, Q, R be arbitrary points on the sides BC, CA, AB of a triangle ABC . Show that the circumcircles of triangles ARQ, BPR , and CQP pass through a common point.

⁶Auguste Miquel was a French mathematician active in the mid-nineteenth century.

Proof. Denote by M the second intersection of the circumcircles of triangles BPR and CQP and suppose it lies inside triangle ABC .



Since quadrilaterals $BPMR$ and $CQMP$ are cyclic, we have

$$\begin{aligned}\angle MRA &= 180^\circ - \angle BRM = \angle MPB = 180^\circ - \angle CPM = \angle MQC \\ &= 180^\circ - \angle AQM.\end{aligned}$$

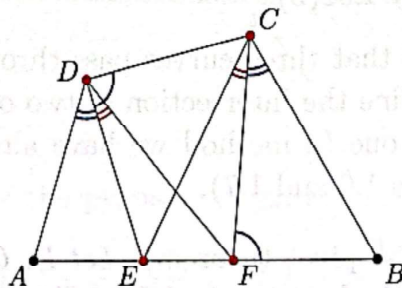
Hence in quadrilateral $ARMQ$ one pair of opposite angles adds up to 180° , so the quadrilateral is cyclic too and the result follows.

The cases when M does not lie inside triangle ABC are treated similarly. $\square$

As we have seen, cyclic quadrilaterals in some sense “multiply” information about angles. Therefore it is worthwhile to look for them when direct angle-chasing seems hopeless.

Example 1.10 (All-Russian Olympiad 1996). Points E and F are chosen on the side AB of a convex quadrilateral $ABCD$ such that $AE < AF$. Given that $\angle ADE = \angle FCB$ and $\angle EDF = \angle ECF$, prove that $\angle FDB = \angle ACE$.

Proof. From $\angle EDF = \angle ECF$ we immediately infer that the quadrilateral $CDEF$ is cyclic.



Hence $\angle BFC = \angle EDC$ and

$$180^\circ - \angle CBA = \angle BFC + \angle FCB = \angle EDC + \angle ADE = \angle ADC.$$

The quadrilateral $ABCD$ is then also cyclic. Thus $\angle ADB = \angle ACB$ and the result follows after subtracting $\angle ADF = \angle ECB$. $\square$

Tangents

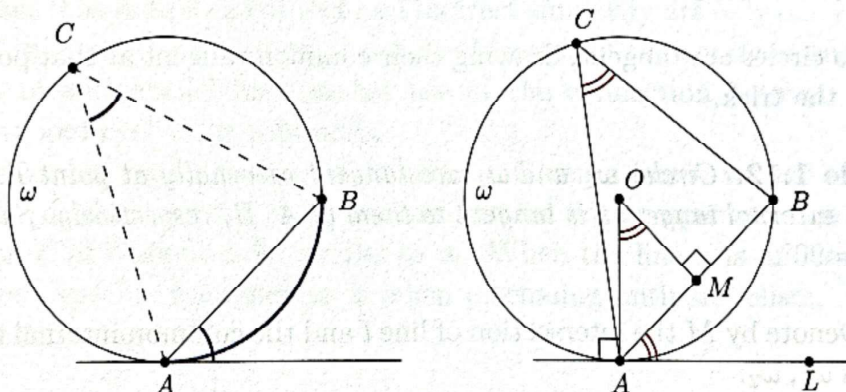
Another phenomenon that can be characterized by angles is tangency.

The crucial result concerning tangents says that the angle between chord AB of a circle and the tangent at A equals the inscribed angle corresponding to arc AB . We formalize it in the following proposition.

Proposition 1.34. *Let ABC be a triangle inscribed in a circle ω . Let ℓ be a line passing through A different from AB . Let L be a point on ℓ such that AB separates points C, L . Then AL is tangent to ω if and only if $\angle LAB = \angle ACB$.*

Proof. Clearly there is only one line passing through A tangent to ω and there is only one line passing through A that satisfies $\angle LAB = \angle ACB$. Hence it is enough to prove that if AL is tangent to ω , then $\angle LAB = \angle ACB$.

Denote by O the center of ω and by M the midpoint of AB .



Since AL is tangent to ω , it is perpendicular to the radius OA . This implies

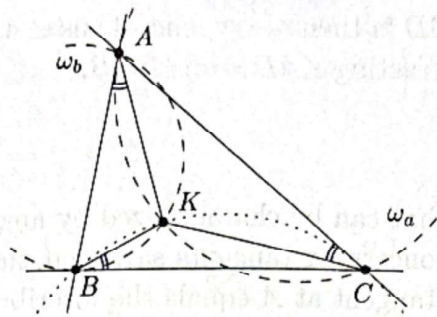
$$\angle LAB = 90^\circ - \angle MAO = \angle AOM = \frac{1}{2} \angle AOB = \angle ACB,$$

and we are done. $\square$

Example 1.11. *Let ABC be a triangle. Denote by ω_a the circle tangent to AB at A and passing through C . Similarly, denote by ω_b the circle tangent to BC at B and passing through A , and ω_c the circle tangent to CA at C and passing through B . Prove that the circles $\omega_a, \omega_b, \omega_c$ intersect at one point.*

Proof. Denote by K the second intersection of circles ω_a and ω_b .

Since BC is tangent to ω_b at B , we get $\angle CBK = \angle BAK$. Similarly, looking at circle ω_a and its tangent AB we obtain $\angle BAK = \angle ACK$. Putting together gives $\angle CBK = \angle ACK$.



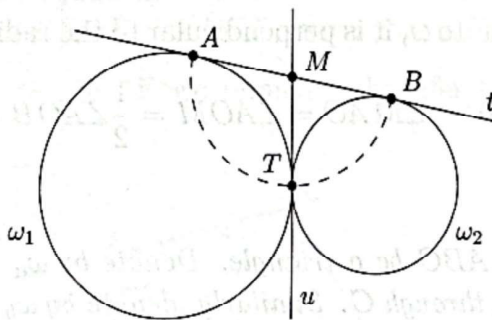
But this implies that CA is tangent to the circumcircle of triangle BCK . Since there is a unique circle passing through B and tangent to AC at C , we get that the circumcircle of BCK is in fact ω_c . So K lies on ω_c and the result follows. $\square$

The point K is called the first Brocard⁷ point of a triangle ABC . The second Brocard point is obtained by intersecting circles defined by reversed order of letters A, B, C .

If two circles are tangent, drawing their common tangent at that point can often do the trick.

Example 1.12. Circles ω_1 and ω_2 are tangent externally at point T . Their common external tangent t is tangent to them at A, B , respectively. Show that $\angle ATB = 90^\circ$.

Proof. Denote by M the intersection of line t and the common internal tangent of circles ω_1, ω_2 .



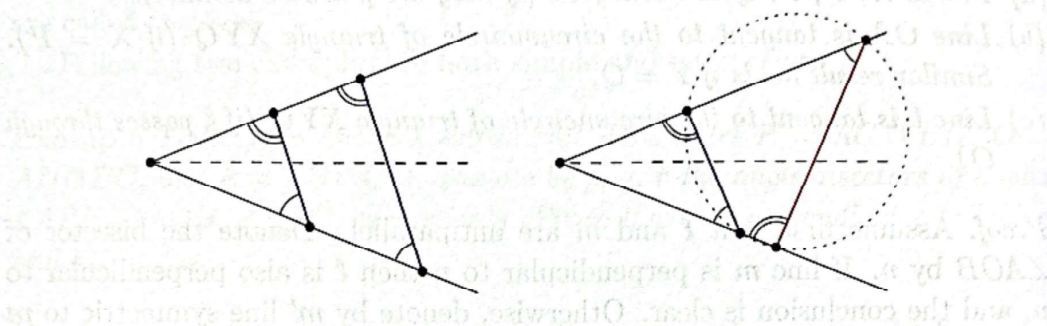
Tangents from M to circles ω_1, ω_2 are equal, so we have $MA = MT = MB$. Thus M is the midpoint of AB and also the circumcenter of triangle ABT implying $\angle BTA = 90^\circ$ as desired. $\square$

⁷Pierre René Jean Baptiste Henri Brocard (1845–1922) was a French mathematician and meteorologist. He is regarded to be one of the co-founders of modern triangle geometry.

Antiparallel lines

In this section we shall discuss another angle-chasing technique. Although this part may seem complicated without bringing immediate reward, the insight it can give is invaluable. We strongly encourage the reader to pay close attention.

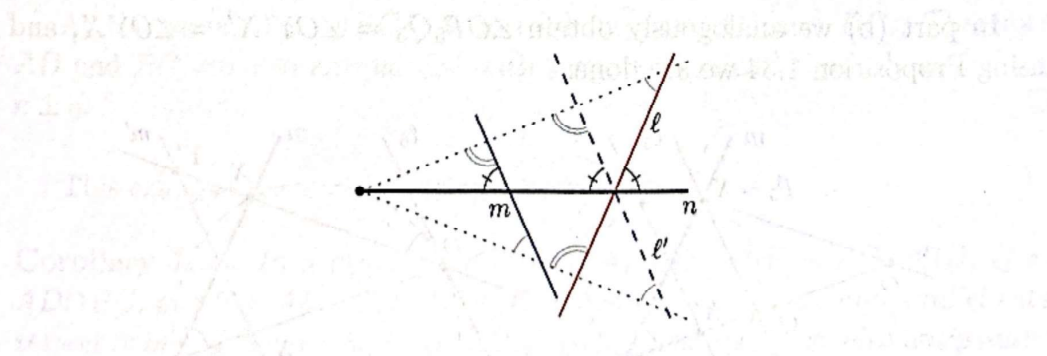
When two similar triangles share one angle, the corresponding sides opposite to the common angle may be either parallel, if the similarity is direct, or they may form a cyclic quadrilateral as we have seen in Corollary 1.30, if the similarity is indirect.



We see that in this case direct and indirect similarity are only one reflection “apart”, namely the reflection about the bisector of the common angle. The concept of antiparallel lines makes use of the connection between indirect similarity and cyclic quadrilaterals.

Now we are ready to form a definition. Given a line n we say that lines ℓ and m (neither parallel to n) are antiparallel with respect to line n if the reflection ℓ' of ℓ about n is parallel to m . When the line n is understood we often omit specific reference to it when discussing antiparallelism. Observe that the following holds:

- (a) If ℓ is antiparallel to m then it is antiparallel to all lines parallel to m .
- (b) (Symmetry) If ℓ is antiparallel to m then m is antiparallel to ℓ .
- (c) Given a line n and a set of mutually parallel lines, then lines antiparallel to all of these form again a set of mutually parallel lines.



The idea should now be clear, it remains to formalize it and describe all

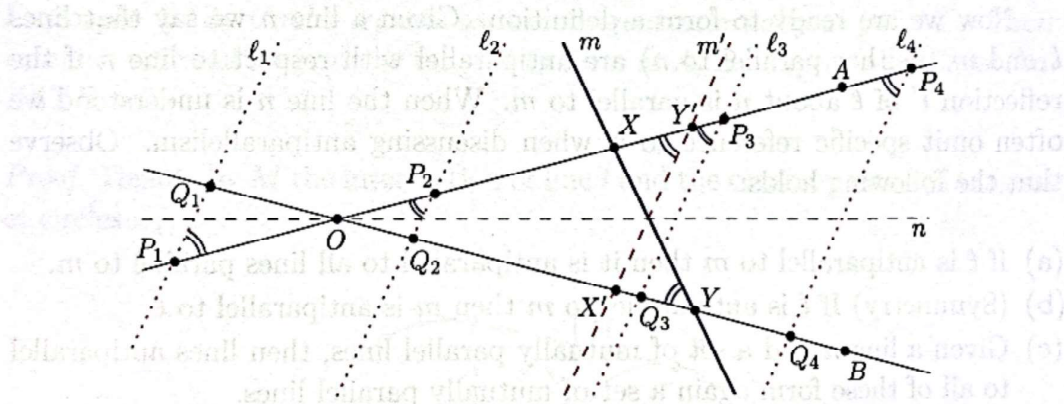
possible cases. As we will see, tangency (as a limit case of concyclicity) will also be involved.

Proposition 1.35. *Let line m intersect ray OA , OB of angle AOB at distinct points X , Y , respectively. Let line ℓ , ($\ell \neq m$) intersect lines OA , OB of angle AOB at (not necessarily distinct) points P , Q , respectively. Then ℓ and m are antiparallel with respect to the angle bisector of angle AOB if and only if one of the following (based on the configuration) holds:*

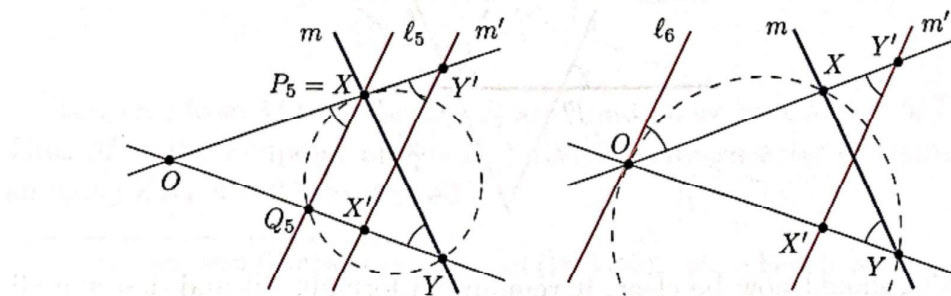
- (a) *Points X , Y , P , Q are concyclic (if they are pairwise distinct).*
- (b) *Line OA is tangent to the circumcircle of triangle XYQ (if $X = P$). Similar result holds if $Y = Q$.*
- (c) *Line ℓ is tangent to the circumcircle of triangle XYO (if ℓ passes through O).*

Proof. Assume first that ℓ and m are antiparallel. Denote the bisector of $\angle AOB$ by n . If line m is perpendicular to n then ℓ is also perpendicular to n , and the conclusion is clear. Otherwise, denote by m' line symmetric to m with respect to n , and by X' , Y' its intersections with OB , OA , respectively.

For part (a), there are four cases to consider corresponding to choices ℓ_1 to ℓ_4 , and P_1 , Q_1 to P_4 , Q_4 (see diagram). In each of them we get $\angle OPQ = \angle OY'X' = \angle OYX$, and the proposition holds.



In part (b) we analogously obtain $\angle OP_5Q_5 = \angle OY'X' = \angle OYX$, and using Proposition 1.34 we are done.



Part (c) is proved by Proposition 1.34 as well.

The converse is proved in the same vein. $\square$

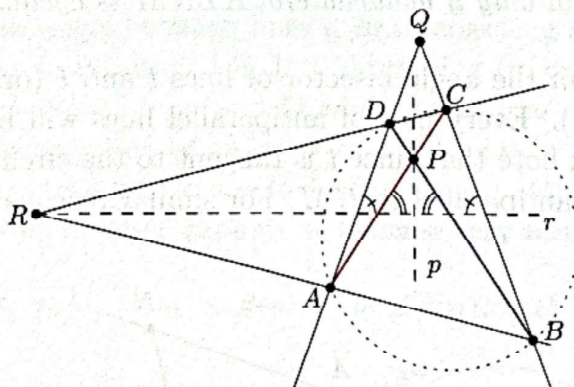
Note that if we work with a stripe and its axis instead of angle AOB with its bisector, the previous proposition is valid trivially.

Since antiparallel lines are usually taken with respect to the angle bisector of some angle, let us in that case call these lines *antiparallel with respect to that angle* or simply *antiparallel in that angle*. Of particular interest are antiparallel lines that both pass through the vertex of an angle – such lines are called *isogonal*.

Following two examples are both simple and instructive.

Example 1.13. In a cyclic quadrilateral $ABCD$ let $P = AC \cap BD$, $Q = AD \cap BC$, and $R = AB \cap CD$. Denote by p , q , r the angle bisectors of angles $\angle APB$, $\angle AQB$, $\angle BRC$, respectively. Prove that r is perpendicular to both p and q .

Proof. We may assume r is horizontal. Since $ABCD$ is cyclic, the lines AC and BD are antiparallel with respect to r , so they form with r an isosceles triangle. In an isosceles triangle with horizontal base, the angle bisector is vertical. Hence $r \perp p$.

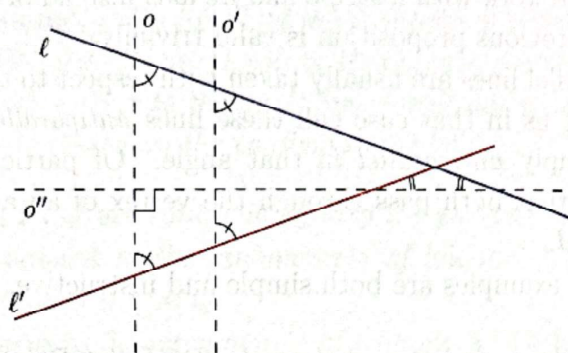


At the same time, $ABCD$ is formed by lines AD and BC . Therefore AD and BC are also antiparallel with respect to r , and similarly as before, $r \perp q$. $\square$

This example has an interesting consequence.

Corollary 1.36. In a cyclic quadrilateral $ABCD$, let $P = AC \cap BD$, $Q = AD \cap BC$, and $R = AB \cap CD$. Let ℓ , ℓ' be two lines which are antiparallel with respect to one of the angles APB , AQB , BRC . Then they are also antiparallel with respect to the remaining two angles.

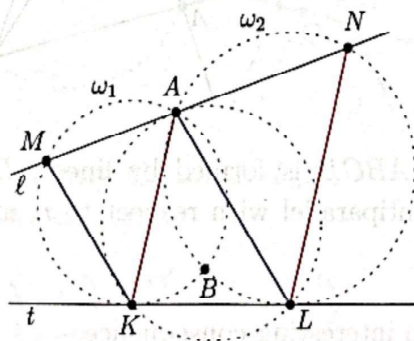
Proof. Clearly, if the lines ℓ, ℓ' are antiparallel with respect to some line o then they are antiparallel with respect to any line o' parallel to o and also with respect to any line o'' perpendicular to o .



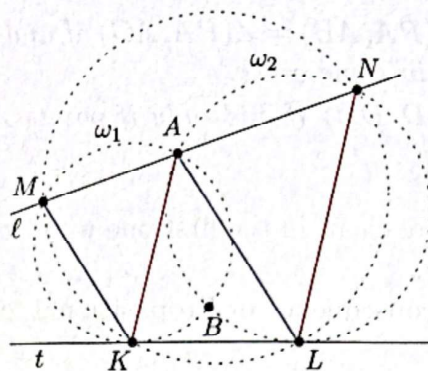
Since any pair of the bisectors of the angles APB, AQB, BRC is either parallel or perpendicular, the conclusion follows. $\square$

Example 1.14 (Czech and Slovak 2010). Circles ω_1, ω_2 intersect at A, B . Their common external tangent t is tangent to them at K, L , respectively, such that B lies inside triangle KLA . Line ℓ passing through A intersects circles ω_1, ω_2 at M, N , respectively. Prove that ℓ is tangent to the circumcircle of triangle KLA if and only if quadrilateral $KLNM$ is cyclic.

Proof. Denote by m the angle bisector of lines t and ℓ (or axis of the stripe if they are parallel). Every pair of antiparallel lines will be considered with respect to m . First note that since t is tangent to the circumcircle of triangle KAM , line KA is antiparallel to KM . For similar reasons is LA antiparallel to LN .



For the if part, if ℓ is tangent to the circumcircle of triangle KLA then KA and AL are antiparallel. Since KM is antiparallel to KA , KA to AL , and AL to LN , together we get that KM is antiparallel to LN . Hence $KLNM$ is cyclic.



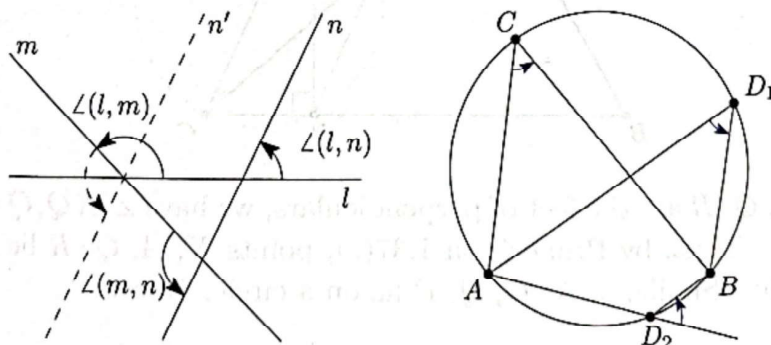
On the other hand, if $KLNM$ is cyclic, AK is antiparallel to KM , KM to LN , and LN to LA . Thus, KA is antiparallel to LA and ℓ is tangent to the circumcircle of triangle KAL . $\square$

Directed angles mod⁸ 180°

We end this section by introducing another advanced concept. Some angle-chasing solutions require casework according to the relative position of points (e.g. Is a triangle acute or obtuse? In which half-plane does an intersection lie? In which order do points lie on a line/circle?). This casework can often be shortened if one uses what is called *directed angles mod 180°* .

Magnitude of an angle between lines l, m intersecting at vertex O can be viewed as a number from interval $[0, 180)$ describing (in degrees) the amount of counter-clockwise rotation around O which takes l to m . Let us call this quantity *the directed measure of an angle* and denote it by $\angle(l, m)$. Note that order of lines in brackets matters – in fact $\angle(l, m) + \angle(m, l) = 180^\circ$. If we adopt this point of view, some properties become very neat.

Proposition 1.37. (a) $\angle(l, m) + \angle(m, n) = \angle(l, n)$, with addition mod 180° .



⁸This means, we shall work with remainders after division by 180. For example, instead of 200° , we shall work with 20° .

- (b) For any point P $\angle(PA, AB) = \angle(PA, AC)$ if and only if points A, B, C lie on a single line in some order.
- (c) $\angle(AC, CB) = \angle(AD, DB)$ if and only if points A, B, C, D lie on one circle in some order.

Proof. First two parts are clear, in the first one we just mustn't forget to work mod 180° (see diagram).

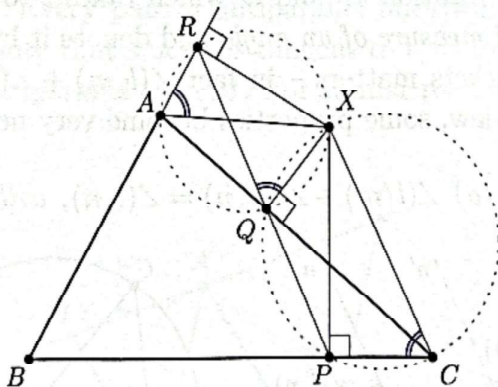
The third part is a consequence of Proposition 1.29. $\square$

Especially, the characterization of cyclic quadrilaterals is very useful. We will demonstrate the use of directed angles on one example, where they simplify the casework substantially (check yourself!).

Example 1.15 (Simson⁹ line). Let ABC be a triangle and X a point in its plane. Denote by P, Q, R the feet of perpendiculars dropped from X to the lines BC, CA, AB , respectively. Prove that P, Q, R lie on a single line if and only if X lies on the circumcircle of the triangle ABC .

Proof. First, assume one of the feet coincides with some vertex of triangle ABC , say P with C .

As Q belongs to AC , the feet P, Q, R are collinear if and only if Q coincides with C or R coincides with A . The first case corresponds to $X = C$, the second to X being antipodal to B , either way the proposition holds. Now let P, Q, R, A, B, C be pairwise distinct.



Since P, Q, R are the feet of perpendiculars, we have $\angle(XQ, QA) = 90^\circ = \angle(XR, RA)$. Thus, by Proposition 1.37(c), points X, A, Q, R lie on a circle in some order. Similarly, X, C, Q, P lie on a circle. Hence

$$\angle(XQ, QR) = \angle(XA, AR) = \angle(XA, AB)$$

⁹Robert Simson (1687–1768) was a Scottish mathematician and professor of mathematics at the University of Glasgow.

and

$$\angle(XQ, QP) = \angle(XC, CP) = \angle(XC, CB).$$

Therefore $\angle(XQ, QR) = \angle(XQ, QP)$ holds if and only if $\angle(XA, AB) = \angle(XC, CB)$ does. The former is equivalent to P, Q, R being collinear, the latter to A, B, C, X being concyclic. The result follows. $\square$

For additional practice give new proofs which avoid casework to Theorem 1.2 and to Theorem 1.33 with points P, Q, R on the sidelines of triangle ABC (not necessarily on the triangle sides).

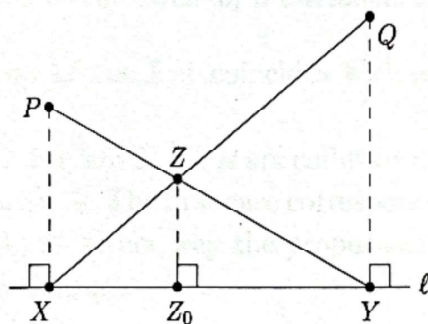
Ratios

Let's start with two examples!

Example 1.16. Let P and Q be distinct points on the same side of line ℓ and let X and Y , respectively, be their projections onto ℓ . Denote by Z the intersection of lines YP and XQ . If $PX = 4$ and $QY = 6$, find the distance from Z to line ℓ .

Proof. Let Z_0 be the projection of Z onto line ℓ . Since $PX \parallel QY$, we see that $\triangle PZX \sim \triangle YZQ$ with factor $k = QY/PX = \frac{3}{2}$. Also, as $ZZ_0 \parallel QY$, we have $\triangle XZZ_0 \sim \triangle XQY$ and we also can find the factor of similarity, since

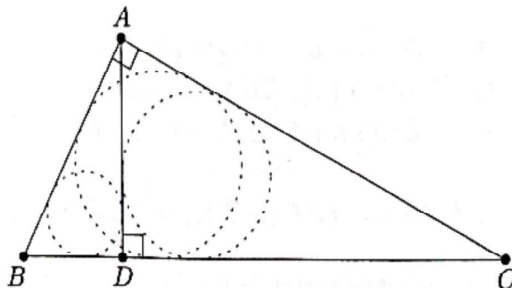
$$\frac{QX}{ZX} = 1 + \frac{ZQ}{ZX} = 1 + k = \frac{5}{2}.$$



From here, we easily deduce that $ZZ_0 = \frac{2}{5}QY = \frac{12}{5}$. □

Example 1.17. Let ABC be a right triangle with $\angle A = 90^\circ$ and AD ($D \in BC$) its altitude. Denote by r , r_1 , r_2 the inradii of triangles ABC , ABD , ACD , respectively. Prove that

$$r^2 = r_1^2 + r_2^2.$$



Proof. Triangles ABC , DBA , DAC are similar (AA), meaning that they are also proportional. In particular, the ratio of inradius over hypotenuse is the same number k for all three of them. Therefore we have

$$r = k \cdot BC, \quad r_1 = k \cdot AB, \quad r_2 = k \cdot AC.$$

Hence we are in fact proving

$$k^2 \cdot BC^2 = k^2 \cdot AB^2 + k^2 \cdot AC^2,$$

which is just the Pythagorean Theorem in triangle ABC . $\square$

We have seen how working with ratios via similarities may lead to solution. Now we are going to develop strong connections between ratios and basic geometric concepts such as concyclicity, collinearity and concurrence.

Power of a Point

Proposition 1.38. (a) Let $ABCD$ be a convex quadrilateral and let $P = AC \cap BD$. Then the points A, B, C, D are concyclic if and only if

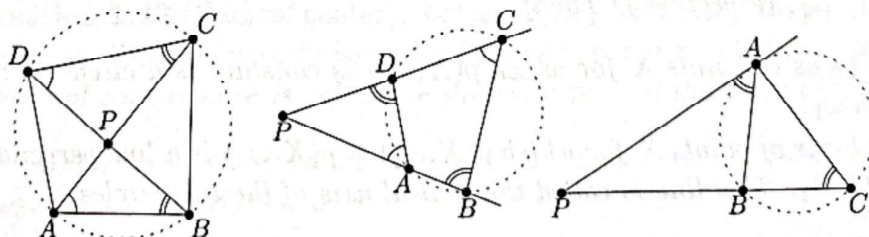
$$PC \cdot PA = PB \cdot PD.$$

(b) Let $ABCD$ be a convex quadrilateral and let $P = AB \cap CD$. Then the points A, B, C, D are concyclic if and only if

$$PA \cdot PB = PC \cdot PD.$$

(c) Assume points P, B, C are collinear in this order and point A does not lie on this line. Then the line PA is tangent to the circumcircle of triangle ABC if and only if

$$PA^2 = PB \cdot PC.$$



Proof. For part (a), observe that the metric condition can be rewritten as $PA/PB = PD/PC$ and is therefore equivalent to similarity of triangles PAB and PDC (SAS). On the other hand, concyclicity of A, B, C, D is by AA also

equivalent to this similarity (see Propositions 1.29 and 1.30(a)), which finishes the proof of (a).

Part (b) is proved in the same fashion using similarity of triangles PAD and PCB .

For (c), line PA is tangent to the circumcircle of triangle ABC if and only if $\angle ACB = \angle PAB$ (recall Proposition 1.34). This is equivalent to $\triangle PAB \sim \triangle PCA$ (AA). Just like in part (a) we deduce that the metric condition is also equivalent to this similarity and we may conclude. $\square$

Thanks to this proposition we can numerically describe the relationship between a circle and a point.

Theorem 1.39 (Power of a Point). *Given point P and circle ω , let ℓ be an arbitrary line passing through P and intersecting ω at points A and B . Then the value of $PA \cdot PB$ does not depend on the choice of ℓ . Also, if P lies outside of ω and PT , $T \in \omega$, is a tangent to ω then $PA \cdot PB = PT^2$.*

If we denote the center of ω by O and its radius by R then

$$PA \cdot PB = |OP^2 - R^2|.$$

The quantity

$$p(P, \omega) = OP^2 - R^2$$

is called the power of point P with respect to circle ω .

Proof. The first part is a direct consequence of the previous proposition.

Formula $PA \cdot PB = |OP^2 - R^2|$ follows if we let ℓ pass through O since then we obtain $PA \cdot PB = (OP + R)(OP - R) = |OP^2 - R^2|$. $\square$

Note that the number $p(P, \omega)$ is negative when P lies inside ω , zero when it lies on ω , and positive otherwise.

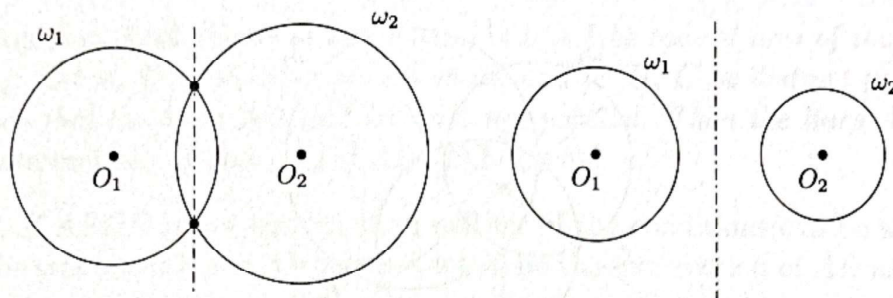
Now let's reveal how fundamental the concept of the Power of a Point is.

Proposition 1.40. *Let ω_1, ω_2 be two circles with distinct centers O_1, O_2 and radii R_1, R_2 , respectively. Then:*

- (a) *the locus of points X for which $p(X, \omega_1)$ is constant is a circle concentric with ω_1 .*
- (b) *the locus of points X for which $p(X, \omega_1) = p(X, \omega_2)$ is a line perpendicular to O_1O_2 . This line is called the radical axis of the two circles.*

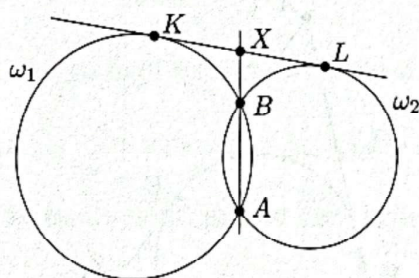
Proof. Part (a) is simple. As $p(X, \omega_1) = XO_1^2 - R_1^2$, we only need XO_1 to be constant. So the locus is indeed a concentric circle (possibly degenerate).

For part (b), we can rewrite the condition as $XO_1^2 - XO_2^2 = R_1^2 - R_2^2$. By the perpendicularity criterion (see Proposition 1.22) such points form a line perpendicular to O_1O_2 . $\square$



The radical axis is a powerful tool in many problems involving intersecting circles since in that case the radical axis is the line joining their intersections, which both have equal (namely zero) power with respect to the two circles.

Proposition 1.41. *Let the circles ω_1, ω_2 intersect at points A, B . Denote by K, L the points of tangency of the common external tangent with circles ω_1, ω_2 , respectively. Then the line AB bisects the segment KL .*



Proof. Let X be the intersection of AB and KL . We know that AB is the radical axis of ω_1 and ω_2 , so we may just write

$$XK^2 = p(X, \omega_1) = p(X, \omega_2) = XL^2,$$

and X is the midpoint of KL . □

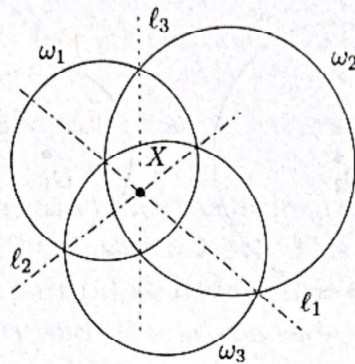
Proposition 1.42 (Radical center). *Let $\omega_1, \omega_2, \omega_3$ be circles with pairwise distinct centers. Then their pairwise radical axes are either parallel or concurrent. The point of concurrence is called the radical center of the three circles.*

Proof. Assume the radical axis ℓ_1 of circles ω_2, ω_3 intersects the radical axis ℓ_2 of ω_3, ω_1 at point X . Then

$$p(X, \omega_2) = p(X, \omega_3) = p(X, \omega_1),$$

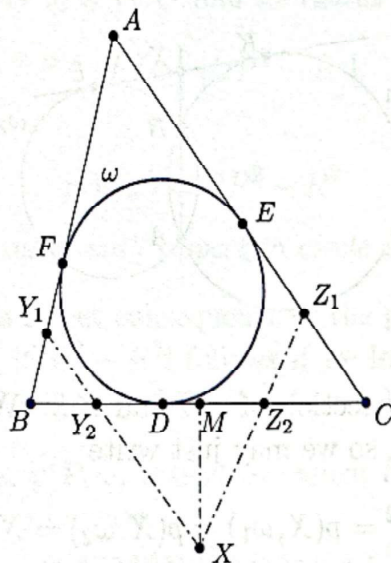
so X lies on the radical axis ℓ_3 of ω_1, ω_2 . □

Next we will see a tricky application of this proposition.



Example 1.18. Let the incircle ω of triangle ABC touch BC , CA , and AB at D , E , and F , respectively. Let Y_1 , Y_2 , Z_1 , Z_2 , and M be the midpoints of BF , BD , CE , CD , and BC , respectively. Let $Y_1Y_2 \cap Z_1Z_2 = X$. Prove that $MX \perp BC$.

Proof. Consider points B and C as circles with zero radii.



Then

$$p(Y_1, \omega) = Y_1F^2 = Y_1B^2 = p(Y_1, B)$$

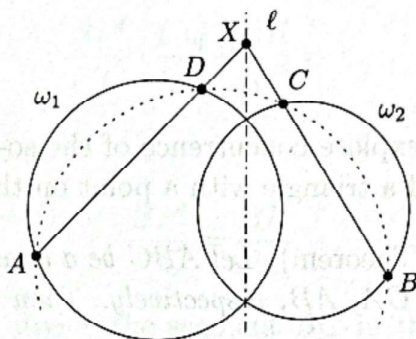
and Y_1 lies on the radical axis of ω and B . Similarly, Y_2 lies on this radical axis. Hence the radical axis of ω and B is precisely line Y_1Y_2 .

Applying the same argument yields that Z_1Z_2 is the radical axis of ω and C . Thus, X is the radical center of the three circles and lies on the radical axis of B and C , i.e. the perpendicular bisector of BC . $\square$

Now we introduce a more applicable form of the previous theorem, which is the heart of many olympiad problems, since it allows us to translate concurrence to concyclicity and vice versa. Further on, we will refer to it as the *Radical Lemma*.

Proposition 1.43 (Radical Lemma). *Let line ℓ be radical axis of the circles ω_1, ω_2 . Let A, D be distinct points on ω_1 and let B, C be distinct points on ω_2 such that the lines AD and BC are not parallel. Then the lines AD and BC intersect at ℓ if and only if $ABCD$ is cyclic.*

Proof. If $ABCD$ is not convex than neither of the conditions can be satisfied and the statement holds. Otherwise, let X be the intersection of AD and BC .



Note that X lies on the radical axis if and only if

$$p(X, \omega_1) = p(X, \omega_2), \quad \text{or equivalently} \quad XD \cdot XA = XC \cdot XB.$$

But the last condition is equivalent to the concyclicity of A, B, C, D (see Proposition 1.38). $\square$

Example 1.19 (IMO 1995). *Let A, B, C, D be four distinct points on a line, in that order. The circles with diameters AC and BD intersect at X and Y . Let P be a point on the line XY such that $P \notin BC$. The line CP intersects the circle with diameter AC at C and M , and the line BP intersects the circle with diameter BD at B and N . Prove that the lines AM, DN, XY are concurrent.*

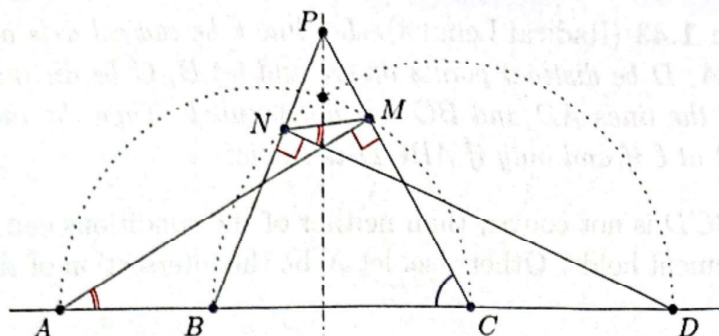
Proof. First assume that P lies outside the circles. Line XY is the radical axis of the two circles, so the concurrence of BN, CM , and XY by the Radical Lemma implies that points B, C, M, N are concyclic. Using the Radical Lemma in the second direction, we realize it suffices to prove that points A, D, M, N are also concyclic.

But this is easy! We just recall that AC and BD are diameters, write

$$\angle DNM + 90^\circ = \angle BNM = 180^\circ - \angle MCB = \angle DAM + 90^\circ,$$

and conclude that $ADMN$ is cyclic.

The case when P belongs to segment XY is treated similarly. Another option is to regard angles as directed mod 180° . $\square$



Ceva's¹⁰ Theorem

In this section we shall explore concurrence of the so-called cevians, the segments joining a vertex of a triangle with a point on the opposite side.

Theorem 1.44 (Ceva's Theorem). *Let ABC be a triangle, and let P , Q , R be points on the sides BC , CA , AB , respectively. Then the following assertions are equivalent:*

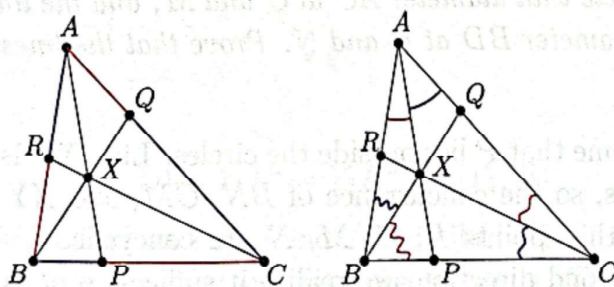
(a) *Lines AP , BQ , CR are concurrent.*

(b)

$$\frac{BP}{PC} \cdot \frac{CQ}{QA} \cdot \frac{AR}{RB} = 1 \quad (*)$$

(c) *(trigonometric form)*

$$\frac{\sin \angle PAC}{\sin \angle BAP} \cdot \frac{\sin \angle QBA}{\sin \angle CBQ} \cdot \frac{\sin \angle RCB}{\sin \angle ACR} = 1$$



Proof. (a) $\Rightarrow$ (b): Assume the cevians are concurrent at X . Now use Area Lemma (see Proposition 1.27) to learn that

$$\frac{[AXB]}{[AXC]} = \frac{BP}{PC}.$$

¹⁰Giovanni Ceva (1647–1734) was an Italian mathematician.

Analogously, we obtain

$$\frac{[BXC]}{[AXB]} = \frac{CQ}{QA} \quad \text{and} \quad \frac{[AXC]}{[BXC]} = \frac{AR}{RB}.$$

Multiplying the three relations gives (*).

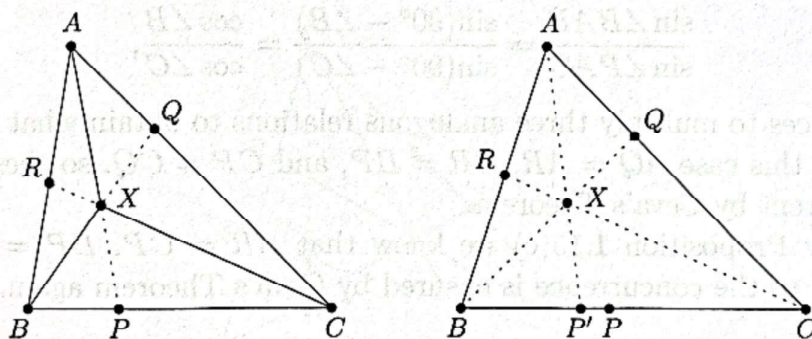
(b) $\Rightarrow$ (a): Assume (*) holds. Intersect BQ and CR at X and also AX and BC at P' . Then AP' , BQ , and CR are concurrent cevians, thus

$$\frac{BP'}{P'C} \cdot \frac{CQ}{QA} \cdot \frac{AR}{RB} = 1.$$

Comparing this with (*) gives

$$\frac{BP'}{P'C} = \frac{BP}{PC},$$

so the points P and P' divide the segment BC in the same ratio and hence they coincide, implying that AP , BQ , CR are concurrent.



(c) $\Leftrightarrow$ (b): Use the Ratio Lemma (see Proposition 1.18) for adjacent triangles ABP and APC to get

$$\frac{BP}{PC} = \frac{AB \sin \angle BAP}{AC \sin \angle PAC}.$$

Analogously, we obtain

$$\frac{CQ}{QA} = \frac{BC \sin \angle CBQ}{AB \sin \angle QBA} \quad \text{and} \quad \frac{AR}{RB} = \frac{AC \sin \angle ACR}{BC \sin \angle RCB}.$$

After multiplying the three equations, simplifying gives

$$\frac{BP}{PC} \cdot \frac{CQ}{QA} \cdot \frac{AR}{RB} = \frac{\sin \angle BAP}{\sin \angle PAC} \cdot \frac{\sin \angle CBQ}{\sin \angle QBA} \cdot \frac{\sin \angle ACR}{\sin \angle RCB},$$

so the two statements are indeed equivalent. $\square$

Ceva's Theorem establishes the existence of many important triangle centers, some of which we have already met.

Corollary 1.45. *In triangle ABC the following cevians (always denote their intersections with BC , CA , AB by P , Q , R , respectively) are concurrent:*

- (a) medians,
- (b) angle bisectors,
- (c) altitudes,
- (d) cevians corresponding to the points of tangency of the incircle (Gergonne¹¹ point),
- (e) cevians corresponding to the points of tangency of the excircles with the triangle sides (Nagel¹² point),

Proof. (a): As $BP = CP$, $AQ = CQ$, and $AR = BR$, concurrence follows from Ceva's Theorem.

(b): We have $\angle BAP = \angle PAC$, $\angle CBQ = \angle QBA$, and $\angle ACR = \angle RCB$, so the result follows from trigonometric form of Ceva's Theorem.

(c): For altitudes we also use the trigonometric form. We have

$$\frac{\sin \angle BAP}{\sin \angle PAC} = \frac{\sin(90^\circ - \angle B)}{\sin(90^\circ - \angle C)} = \frac{\cos \angle B}{\cos \angle C},$$

and it suffices to multiply three analogous relations to obtain what we need.

(d): In this case $AQ = AR$, $BR = BP$, and $CP = CQ$, so these cevians are concurrent by Ceva's Theorem.

(e): By Proposition 1.15(c) we know that $AR = CP$, $BP = AQ$, and $CQ = BR$, so the concurrence is ensured by Ceva's Theorem again. $\square$

Example 1.20. *Points M , N on the sides AB , AC of the triangle ABC satisfy $MN \parallel BC$. Prove that lines BN and CM intersect on the A -median of triangle ABC .*

Proof. Since $MN \parallel BC$, the sides AB , AC are divided by M , N , respectively, in the same ratio. In other words,

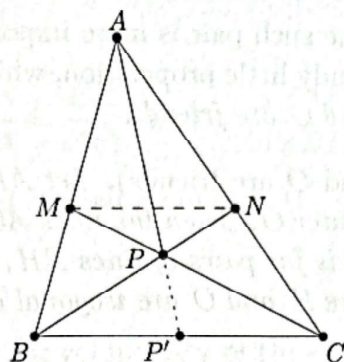
$$\frac{AM}{MB} = \frac{AN}{NC}.$$

Denote by P the intersection of BN and CM and by P' the intersection of AP and BC . Then Ceva's Theorem for concurrent cevians AP' , BN , CM implies

$$\frac{BP'}{P'C} \cdot \frac{CN}{NA} \cdot \frac{AM}{MB} = 1, \quad \text{hence} \quad \frac{BP'}{P'C} = \frac{AN}{NC} \cdot \frac{MB}{AM} = 1.$$

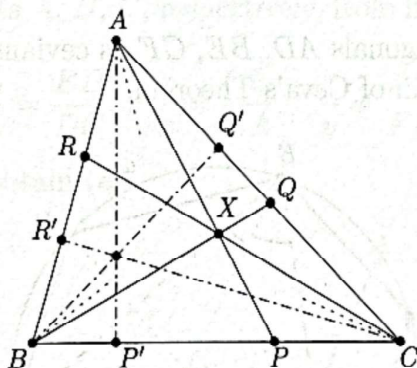
¹¹Joseph Diaz Gergonne (1771–1859) was a French mathematician and logician.

¹²Christian Heinrich von Nagel (1803–1882) was a German geometer.



Hence $BP' = P'C$ and P lies on the A -median. $\square$

Theorem 1.46 (Existence of isogonal conjugate). *Let cevians AP , BQ , CR concur at point X . Now construct cevians AP' , BQ' , CR' which are isogonal to AP , BQ , CR , respectively, in the respective angles. Then the cevians AP' , BQ' , CR' are concurrent. The point of concurrence is called the isogonal conjugate of X .*



Proof. As AP' is isogonal to AP in $\angle A$, we have $\angle BAP = \angle P'AC$ and $\angle PAC = \angle BAP'$, and similarly for the other cevians. In fact,

$$\frac{\sin \angle BAP}{\sin \angle PAC} \cdot \frac{\sin \angle CBQ}{\sin \angle QBA} \cdot \frac{\sin \angle ACR}{\sin \angle RCB} = \frac{\sin \angle P'AC}{\sin \angle BAP'} \cdot \frac{\sin \angle Q'BA}{\sin \angle CBQ'} \cdot \frac{\sin \angle R'CB}{\sin \angle ACR'}.$$

However, by trigonometric form of Ceva's Theorem, the left-hand side of this equation equals 1 as AP , BQ , and CR are concurrent. Hence also the right-hand side equals 1, and AP' , BQ' , and CR' are concurrent too. $\square$

We can easily see that the relation of isogonal conjugation is symmetric and except for the incenter, which is the conjugate of itself, it pairs up the points in the triangle. It should be noted that the concept of isogonal conjugation can be easily extended also to points in the exterior of triangle ABC .

We dare to say that one such pair is more important than others. Details are exposed in the next handy little proposition, which will be referred to with a familiarizing name *H* and *O* are friends.

Proposition 1.47 (*H* and *O* are friends). Let ABC be a triangle with orthocenter H and circumcenter O . Then the lines AO and AH are isogonal in $\angle A$, and similar result holds for pairs of lines BH, BO and CH, CO , in the respective angles. Therefore H and O are isogonal conjugates.

Proof. If triangle ABC is acute, we have simply

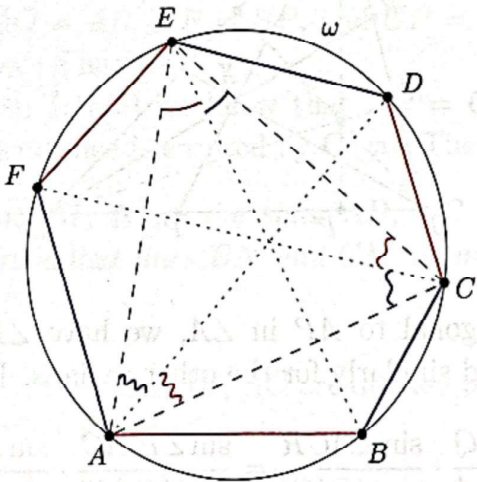
$$\angle BAO = \frac{1}{2}(180^\circ - \angle AOB) = 90^\circ - \angle C = \angle CAH,$$

and the conclusion follows. In the other cases we proceed similarly. $\square$

Example 1.21 (China MO training 1988). Let $ABCDEF$ be a hexagon inscribed in a circle ω . Show that the diagonals AD, BE, CF are concurrent if and only if

$$AB \cdot CD \cdot EF = BC \cdot DE \cdot FA.$$

Proof. Consider the diagonals AD, BE, CF as cevians in triangle ACE , and apply trigonometric form of Ceva's Theorem.



The diagonals are concurrent if and only if

$$\frac{\sin \angle CAD}{\sin \angle DAE} \cdot \frac{\sin \angle ECF}{\sin \angle FCA} \cdot \frac{\sin \angle AEB}{\sin \angle BEC} = 1. \quad (\clubsuit)$$

Now denote by R the radius of ω . The Extended Law of Sines yields

$$\frac{\sin \angle CAD}{\sin \angle DAE} = \frac{\frac{CD}{2R}}{\frac{DE}{2R}} = \frac{CD}{DE}.$$

In a similar fashion, we obtain

$$\frac{\sin \angle ECF}{\sin \angle FCA} = \frac{EF}{AF} \quad \text{and} \quad \frac{\sin \angle AEB}{\sin \angle BEC} = \frac{AB}{BC}.$$

Plugging this into (♣) and expanding implies the result. $\square$

Menelaus'¹³ Theorem

Surprisingly, the criterion for collinearity of three points on the triangle sides (possibly extended) has similar form.

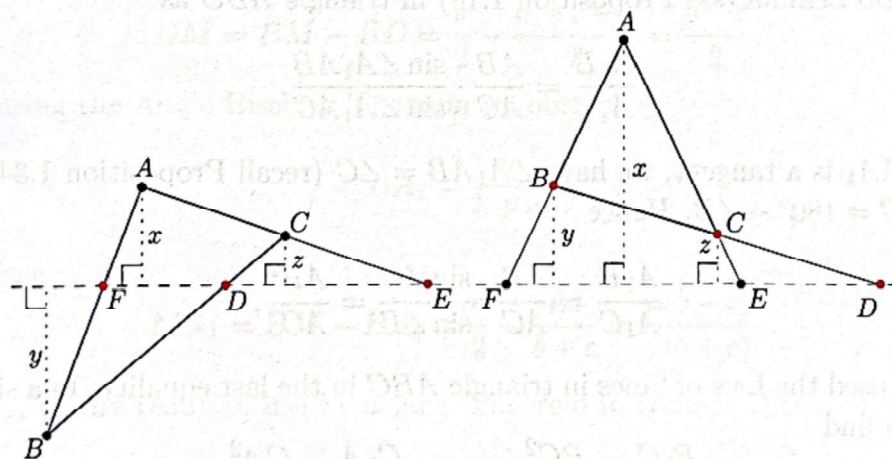
Theorem 1.48 (Menelaus' Theorem). *Let ABC be a triangle and let points D, E, F lie on the lines BC, CA, AB , respectively, so that either none or two of them lie on the triangle sides. Then the points D, E, F are collinear if and only if*

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1. \quad (\spadesuit)$$

Proof. Assume first that D, E, F are collinear on a line ℓ and denote by x, y, z the distances of points A, B, C , respectively, from line ℓ . Now using similar triangles yields

$$\frac{y}{z} = \frac{BD}{DC}, \quad \frac{z}{x} = \frac{CE}{EA}, \quad \frac{x}{y} = \frac{AF}{FB}.$$

Multiplying these we obtain (♠).



Now assume (♠) holds and let D' be the intersection of EF and BC . Points D', E, F are collinear and we may apply the first part of the statement to obtain

$$\frac{BD'}{D'C} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1.$$

¹³Menelaus of Alexandria (c. 70–140) was a Greek mathematician and astronomer.

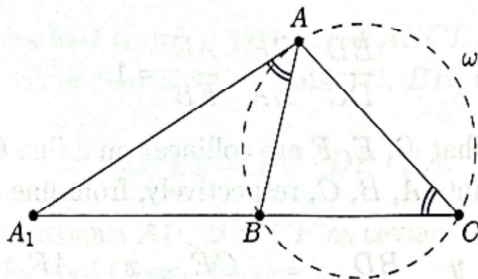
Comparing this with ($\spadesuit$) gives

$$\frac{BD'}{D'C} = \frac{BD}{DC}. \quad (\diamond)$$

Now realize that both D and D' lie either inside segment BC or outside of it. Either way, ($\diamond$) implies that D and D' coincide, so D, E, F are collinear. $\square$

Thanks to Menelaus' Theorem we can sometimes focus only on a small part of a complicated picture.

Example 1.22. Let ω be the circumcircle of triangle ABC and let the tangent to ω at A intersect BC at A_1 . Define points B_1, C_1 analogously. Prove that A_1, B_1, C_1 are collinear.



Proof. In order to use Menelaus' Theorem we first calculate A_1B/A_1C from the Ratio Lemma (see Proposition 1.18) in triangle ABC as

$$\frac{A_1B}{A_1C} = \frac{AB}{AC} \cdot \frac{\sin \angle A_1AB}{\sin \angle A_1AC}.$$

Since AA_1 is a tangent, we have $\angle A_1AB = \angle C$ (recall Proposition 1.34) and $\angle A_1AC = 180^\circ - \angle B$. Hence

$$\frac{A_1B}{A_1C} = \frac{AB}{AC} \cdot \frac{\sin \angle C}{\sin \angle B} = \frac{AB^2}{AC^2},$$

having used the Law of Sines in triangle ABC in the last equality. In a similar vein we find

$$\frac{B_1C}{B_1A} = \frac{BC^2}{BA^2} \quad \text{and} \quad \frac{C_1A}{C_1B} = \frac{CA^2}{CB^2}.$$

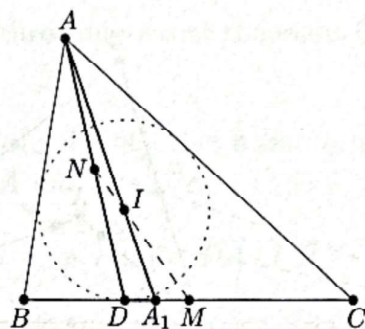
Multiplying the three fractions gives

$$\frac{A_1B}{A_1C} \cdot \frac{B_1C}{B_1A} \cdot \frac{C_1A}{C_1B} = 1,$$

and Menelaus' Theorem implies the result. $\square$

Example 1.23. Let ABC be a scalene triangle and M the midpoint of BC . The incircle centered at I touches BC at D . Denote by N the midpoint of AD . Prove that N, I, M are collinear.

Proof. We may assume $b > c$. Let AA_1 be the internal angle bisector with $A_1 \in BC$. We are going to apply Menelaus' Theorem in triangle ADA_1 .



We know that $DN = AN$ and by Corollary 1.28

$$\frac{A_1I}{IA} = \frac{a}{b+c},$$

so it remains to calculate MD and MA_1 .

Since $2BD = a + c - b$ (see Proposition 1.15(a)) and M is the midpoint of BC , we get

$$DM = BM - BD = \frac{a}{2} - \frac{a+c-b}{2} = \frac{b-c}{2}.$$

Now using the Angle Bisector Theorem we obtain

$$BA_1 = \frac{ac}{b+c},$$

therefore

$$MA_1 = BM - BA_1 = \frac{a}{2} - \frac{ac}{b+c} = \frac{a(b-c)}{2(b+c)}.$$

Finally, we are ready to use Menelaus' Theorem in triangle ADA_1 . Since

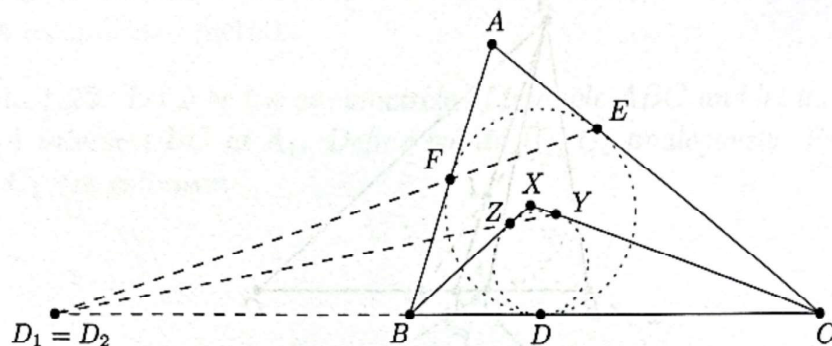
$$\frac{AN}{ND} \cdot \frac{DM}{MA_1} \cdot \frac{A_1I}{IA} = 1 \cdot \frac{\frac{b-c}{2}}{\frac{a(b-c)}{2(b+c)}} \cdot \frac{a}{b+c} = 1,$$

the collinearity of N, I , and M follows. $\square$

The following rather subtle example summarizes all techniques discussed in this section.

Example 1.24 (IMO 1995 shortlist). *The incircle of triangle ABC touches the sides BC , CA , AB at points D , E , F , respectively. Let X be a point inside the triangle ABC such that the incircle of triangle XBC touches BC , CX , XB at D , Y , Z , respectively. Show that E , F , Z , and Y are concyclic.*

Proof. First, if $AB = AC$, then D is the midpoint of BC and triangle XBC is also isosceles. Therefore $ZYEF$ is an isosceles trapezoid, hence circumscribable. Now assume $AB \neq AC$. Since BC is a common tangent of the



two circles, it is also their radical axis. Hence by the Radical Lemma (see Proposition 1.43) it suffices to prove that BC , EF , and YZ are concurrent. Let $D_1 = EF \cap BC$ and $D_2 = YZ \cap BC$. The key idea is to compare Ceva's Theorem (for triangle ABC and concurrent cevians AD , BE , CF – see Corollary 1.45(d)) and Menelaus' Theorem (for triangle ABC and line EF). We obtain

$$\frac{BD}{DC} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB} = 1 = \frac{BD_1}{D_1C} \cdot \frac{CE}{EA} \cdot \frac{AF}{FB}, \quad \text{hence} \quad \frac{BD}{DC} = \frac{BD_1}{D_1C}.$$

Now we use the same technique for triangle XBC , concurrent cevians XD , BY , CZ , and line YZ . We get

$$\frac{BD}{DC} \cdot \frac{CY}{YX} \cdot \frac{XZ}{ZB} = 1 = \frac{BD_2}{D_2C} \cdot \frac{CY}{YX} \cdot \frac{XZ}{ZB}, \quad \text{hence} \quad \frac{BD}{DC} = \frac{BD_2}{D_2C}.$$

Comparing yields

$$\frac{BD_1}{D_1C} = \frac{BD_2}{D_2C}$$

and since neither of D_1 , D_2 lies on segment BC , the points must coincide. Hence BC , EF , and YZ are concurrent and we may conclude. $\square$

Directed segments

Applying Ceva's or Menelaus' Theorem may cause some painful casework as we should be sure about the relative positions of the involved points. However, this may be simplified if we adopt Newton's¹⁴ concept of directed segments.

¹⁴Isaac Newton (1643–1727) was an English physicist, mathematician and natural philosopher.